How accurate LIGHT 1Dv0_2 is

The 1-D model on v0_2's chemistry, version 0.1.2+1d0, on 30 September 2026, with its energy ladder, its system map and its reaction rates

Bottom line. LIGHT 1Dv0_2 runs v0_2's chemistry file for file, and that chemistry reproduces the fitted KrF decays [1, 2] within 2.3 to 6.1 % rms and predicts decays that no fit used [4] within 11.7 % rms. At the Tahoma laser [5] on 1Dv0_2's own power input the centre cell gives 86.5 % of the measured small-signal gain of 6.39 %/cm.

The 1-D layer has no measurement to meet. Its centre cell matches v0_2's 0-D run to 1.7×10^{-5} at 248 nm on 32 to 256 cells, and at Tahoma the gap average on 64 cells gives 84 %, where v0_2 under the beam drive gives 89 %. The cells at the walls are unresolved sheaths, so a gap average approaches the interior only first order in the cell size and is quoted with its mesh. The C/B emission ratio of Yu and Setser [3] is matched to -0.5 %. No fit used Tahoma's numbers and no rate was set by a laser measurement, so the Tahoma scores are predictions.

The chemistry's largest misses are on data that conflict with the fitted sets, 19.3 and 34.1 % rms on the C-A decays of Hakuta et al. [6] and -22.3 % on the KrF(B) lifetime of Burnham and Searles [7], and the VUV yield of pure krypton at low pump, 136 % of Morozov's measurement [8].

[1] Eden, Waynant, Searles and Burnham, Appl. Phys. Lett. 32, 733 (1978). [2] Gadomski, Xu, Setser and Nelson, Chem. Phys. Lett. 189, 153 (1992). [3] Yu and Setser, J. Phys. Chem. 94, 2934 (1990). [4] Xu, Gadomski and Setser, J. Chem. Phys. 99, 2591 (1993). [5] Spectra Technology, *Experimental Support for KrF Laser Modeling*, final report to Los Alamos, 19 July 1989, OSTI 10136715. [6] Hakuta et al., Jpn. J. Appl. Phys. 29, 1194 (1990). [7] Burnham and Searles, J. Chem. Phys. 67, 5967 (1977). [8] Morozov, Heindl, Krücken, Ulrich and Wieser, J. Appl. Phys. 103, 103301 (2008).

LIGHT 1Dv0_2 runs v0_2's chemistry in every cell of a gap on Nathaniel's Fluid1D, which it imports and never edits. Its input is v0_2's, a power deposited in the electrons, uniform across a 1-cm gap between two grounded walls. Fluid1D moves the charged species and the electron energy between the cells and into the walls under the field it solves, and every row of v0_2's network runs in every cell. The chemistry is v0_2's file for file. The 9 files that make it have v0_2's md5 sums, which the generator checks before it writes a number, so every score of the chemistry is v0_2's, taken here from the v0_2 overview. What the 1-D layer adds has no measurement to meet, and its tests are internal. With uniform power the interior must be v0_2's 0-D run, and at v0_2's reference point the centre cell's 248-nm share is v0_2's to 1.7×10^{-5} on 32 to 256 cells. At Tahoma on 1Dv0_2's own power input the centre cell's gain is v0_2's to 2.8×10^{-4}, $\times0.865$ the measured value. The cells next to the walls are unresolved sheaths. They cut 6.22 % of the input on 32 cells and 0.79 % on 256, so a gap average approaches the interior first order in the cell size. The stage computes every rate as v0_2's model does, with one difference above 30 eV (Section 4). Table 1 lists every test.

1 How the numbers were made

Every number in this note is written by `docs/make_overview_1Dv0_2.py`, and those of the chemistry by `docs/make_overview_v0_2.py` and `docs/make_system_map_v0_2.py`, from the models themselves, and a test fails when those files no longer match them. The 1-D script first compares the md5 sums of 1Dv0_2's chemistry, `model.py`, `network.py`, `species.py`, `xsec.py`, `two_level.py`, `data/Kr_Biagi71_LXCat.txt`, `data/F2_SIGLO_LXCat.txt`, `data/kr_tail_ratio.txt`, `report.py`, with v0_2's and writes nothing if one differs. Fluid1D (`nathaniel/1DModelKrF/src`) needs Python 3.11 or later with numba, so the 1-D numbers come from that interpreter, on four numba threads. The Tahoma runs take ML's B/C constants and conditions, as the v0_2 overview's do, with ML bound to v0_2.

1Dv0_2 has no drive, beam or cavity module, which is LIGHT's rule. So it cannot be scored under Tahoma's electron beam, and the beam scores in Table 1 are v0_2's in 0-D. 1Dv0_2 is scored at Tahoma on its own power input (Section 7).

test	measured, or v0_2	1Dv0_2	role
chemistry files against v0_2 0.1.2's, md5		all 9 identical	identity
every rate of the stage against v0_2's model, 800 random cell states		T_g and density rows exactly, T_e tables 3.0×10^{-4}, tail 7.3×10^{-6}	consistency
$Kr_2^+ + e^-$ above 30 eV		up to 17 % high	a difference
centre cell at the reference point, 32 to 256 cells	248 nm 4.66672 %, 147 nm 20.0184 %	to 1.7×10^{-5} and 8.7×10^{-6}, T_g to 2.3×10^{-6}	0-D limit
the same, Kr_2^* + F at the top of its band	9.49835 %	9.49834 %, to 5.8×10^{-7}	0-D limit
centre cell at Tahoma, own input, seeds 10^{13} and 10^{15} cm^{-3}	5.530 and 5.555 %/cm	to 2.8×10^{-4} and 2.0×10^{-4}	0-D limit
energy ledger of the gap, every run		closes to 10^{-12} of the input	bookkeeping
energy to the walls, reference point		0.036 % on 256 cells	transport
cut by the mean-energy bounds in the wall cells		6.22 to 0.79 % of the input	unresolved sheath
gap average's deficit, each doubling of the mesh		falls $\times$2.00 to $\times$2.02	first order in Δx
T_e transient against v0_2, Δt 50 to 12.5 ps		1.3 to 0.4 %	first order in Δt
Tahoma from ML's seed, 10^{10} cm^{-3}		no breakdown, 100.0 % cut	the seed
KrF(B) and KrF(B,C) decays, Eden 1978, Gadomski 1992, Yu and Setser 1990		rms 2.3 to 6.1 %	train
KrF(B,C) decays, Xu 1993 Figs. 5 to 7		rms 11.7 %	held out
C-A decays, Hakuta 1990, and τ_B, Burnham and Searles 1977		rms 19.3 and 34.1 %, and -22.3 %	held out, in conflict
VUV yield of pure Kr at low pump, Morozov 2008	0.42 ± 0.05	0.57, $\times$1.36	check
Tahoma g_0 under the beam drive, v0_2 in 0-D	6.39 %/cm	5.69, $\times$0.89	device, score only
Tahoma I_{sat} and $g_0 I_{sat}/P_{in}$, the same	1.4 and 0.27	$\times$1.11 and $\times$0.99	device, score only
Tahoma g_0 on 1Dv0_2's own input, centre cell	6.39 %/cm	5.528, $\times$0.865	device, score only
the same, gap average		5.36, $\times$0.84	device, score only

Table 1: LIGHT 1Dv0_2 against v0_2 (the first block), and its chemistry against the data (the second, v0_2's own scores, since the files are v0_2's) and at Tahoma (the third). A relative number is 1Dv0_2's deviation from v0_2's 0-D run or model, a percentage in the second block v0_2's deviation from the data, and a $\times$ factor the model over the measurement. The reference point is v0_2's, 1 atm of Kr with 0.5 % F_2 at 300 kW cm^{-3} for 1 μs, run to 2 μs with Δt = 50 ps.

2 The 1-D layer

Each step of Δt runs the chemistry for $\Delta t/2$, Fluid1D's transport for Δt and the chemistry for $\Delta t/2$ again, Fluid1D's source-coupled split. The chemistry stage runs every row of v0_2's network in every cell with the electron energy, in substeps that change no density by more than 5 % (at most 8000 per stage), and reads the rates at each substep's starting state. The stage's energy source is the given power, plus the field's positive work on the electrons, minus v0_2's elastic loss, which heats the gas. Each cell's T_g rises at constant volume, as in v0_2.

Fluid1D brings what v0_2 does not have. It solves the space-charge field between grounded walls and drifts and diffuses the electrons, the ions and the electron energy. The ion mobilities are Fluid1D's own cited table for Kr^+, Kr_2^+ and F^- in Kr. The electron mobility and diffusion are a Maxwellian at the mean energy on 1Dv0_2's own cross sections, and the gas temperature the transport reads is held at the fill's. The neutrals do not move, since 1 μs at 1 atm spreads them by microns. The mean electron energy is held between 0.03 and 60 eV, so T_e between 0.02 and 40 eV.

3 The interior is v0_2's 0-D run

With uniform power, and walls that take 0.036 % of the input, the cells away from the walls see what a 0-D volume sees. At v0_2's reference point the centre cell sends 4.66664 % of its input out at 248 nm on 256 cells, where v0_2 gives 4.66672 %. Over 32 to 256 cells it lies within 1.7×10^{-5} of v0_2 at 248 nm and 8.7×10^{-6} at 147 nm, and its gas ends within 2.3×10^{-6} of v0_2's 649.35 K. On every mesh each cell more than 0.05 cm from the walls sends out the centre cell's 248-nm share to 8.3×10^{-7}.

A band reaches the 1-D chemistry as it reaches v0_2's. The centre cell on 64 cells, with Kr_2^* + F at the top of its band, 1.0×10^{-9} cm^3s^{-1}, gives 9.49834 % at 248 nm against v0_2's 9.49835 % with the same band, and 0.462 % at 147 nm against 0.462 %.

4 The stage's rates

The stage computes every rate on arrays, and a compiled kernel does the same inside the substeps. At 800 random cell states, every species over 7 decades, T_g from 250 to 1500 K and T_e up to 33 eV, with the bands at their defaults and with the three unmeasured ones set, 33,119 rates were compared with v0_2's own `Model._k`. The rows that depend on T_g and the densities agree exactly. The rows read from T_e tables agree to 3.0×10^{-4}, largest just above the 0.02-eV floor of v0_2's $Kr_2^+ + e^-$ power law, where the table interpolates across the law's kink. The two-temperature tail agrees to 7.3×10^{-6}, and the kernel matches the arrays exactly.

There is one difference. The stage tabulates every T_e-dependent row on v0_2's nodes, which end at 30 eV, and holds it there above. v0_2's cross-section tables do the same, but v0_2 evaluates $Kr_2^+ + e^-$ recombination (R11p), Shiu and Biondi's power law, at any T_e. Above 30 eV the stage's rate is therefore higher than v0_2's, by up to 6 % among the random states and by 17 % at the 40-eV ceiling. The drained wall cells sit there, and so does a cell that fails to break down. The interior passes 30 eV only briefly while a low seed breaks down, when few ions exist to recombine. At Tahoma from a seed of 10^{13} cm^{-3}, where v0_2's own T_e peaks at 38 eV, the centre cell still gives v0_2's gain to 2.8×10^{-4}. The random states of 1Dv0_2's own rate test stay below the table's edge, which is why that test passes.

5 The wall cells and the mesh

The walls drain the electrons of the cells next to them, and the given power then heats few electrons there. The mean-energy ceiling cuts what they cannot hold, and Fluid1D's split gives them the field's positive work only and keeps out a phantom negative part. Both shrink as the mesh refines (Table 2). The deficit of the gap average below v0_2's 0-D share falls by $\times 2.00$ to $\times 2.02$ with each doubling of the mesh, first order in the cell size. So a gap average is quoted only with its mesh, and the interior is the model's answer. The walls themselves take 0.036 % of the input on 256 cells. The ledger of every run closes to 10^{-12} of the input, recomputing the gap's chemical, electron and thermal energy from its state.

The cells beside the wall cell do not converge. The first of them sends out 4.668, 4.672, 4.684 and 4.751 % on 32 to 256 cells, moving away from the interior's value as the mesh refines. On 256 cells the cells alternate about the interior value, and n_e and T_e with them (Figure 1), out to 195 μm from the wall before they agree with the centre cell to 10^{-3}. The cells there are 39 μm wide, and the Debye length in the interior at mid-pulse is 1.0 μm, so no mesh here resolves the sheath, and the alternation is the unresolved sheath's, not physics. It largely cancels in the gap average, which still converges.

cells	248 nm, % of the input			cut by the	phantom work	to the	gap T_g
	centre cell	beside the wall cell	gap	bounds (%)	kept out (%)	walls (%)	at the end (K)
32	4.66667	4.668	4.376	6.22	86.41	0.052	628
64	4.66665	4.672	4.521	3.10	16.01	0.042	638
128	4.66665	4.684	4.594	1.54	3.21	0.041	644
256	4.66664	4.751	4.631	0.79	0.84	0.036	647

Table 2: v0_2's reference point across a 1-cm gap on four meshes. v0_2's 0-D run sends 4.66672 % out at 248 nm and ends at 649.35 K.

6 The time step

The split and the stage are first order in Δt. The stage reads the rates at each substep's start, so a steady state is exact, and the error lies in transients. In a 20-ns pulse on 8 cells the centre cell's T_e differs from v0_2's by at most 1.3 % at 50 ps, 0.6 % at 25 ps and 0.4 % at 12.5 ps, and each halving of the step cuts the error by $\times 1.71$ to $\times 2.03$. The reference point runs at 50 ps.

7 Tahoma and the seed

At Tahoma's conditions, 2.8 Torr of F_2 in Kr at 665 Torr, 330 kW cm^{-3} for 450 ns, with ML's constants imposed and g_0 from KrF(B) over 200 to 400 ns on 64 cells, 1Dv0_2's centre cell gives v0_2's own-input gain to 2.8×10^{-4} from a seed of 10^{13} cm^{-3} and to 2.0×10^{-4} from 10^{15} (Table 3). That is $\times 0.865$ the measured 6.39 %/cm. The gap average, lowered by the wall cells, is $\times 0.84$. The power input sets the electron gas apart from the beam's (the v0_2 overview, Section 8), so this gain does not test the input either.

From ML's own seed, 10^{10} cm^{-3}, 1Dv0_2 never breaks down. v0_2's 0-D T_e runs to 73 eV from that seed before its tail reaches the Kr thresholds, and 1Dv0_2 holds T_e below 40 eV, so the ceiling cuts 100.0 % of the input and no KrF forms. From 10^{13} v0_2's 0-D T_e peaks at 38 eV. So 1Dv0_2 needs a seed of 10^{13} cm^{-3} or more at these powers, while v0_2's own gain barely moves with the seed.

seed (cm^{-3})	v0_2, 0-D g_0 (%/cm)	centre cell g_0 (%/cm)	gap average g_0 (%/cm)	cut by the bounds (% of input)
10^{13}	5.530	5.528	5.36	3.1
10^{15}	5.555	5.554	5.38	3.1
10^{10}	5.530	0.000	0.00	100.0

Table 3: Tahoma on the power input with ML's constants, v0_2 in 0-D and 1Dv0_2 on 64 cells. Tahoma measured 6.39 %/cm.

8 The chemistry

The chemistry is v0_2's, and the v0_2 overview gives its accuracy in full. In short, the KrF(B,C) block is a joint fit to the three kinetic measurement sets that agree, matched to 2.3 to 6.1 % rms, and it predicts the held-out decays of Xu et al. to 11.7 %. The KrF(B) radiative lifetime, 6.99 ns, is the one KrF rate neither measured nor fitted. Under Tahoma's beam v0_2 gives g_0 at $\times 0.89$, I_{sat} at $\times 1.11$ and $g_0 I_{sat}/P_{in}$ at $\times 0.99$, and over the B/C posterior g_0 spans 4.99 to 6.18 %/cm. The Kr_2^* route is basic, and the rates known only as a published spread are bands whose default is a measurement or 0. Two bands matter at the reference point, Kr_2^* + F (4.67 to 9.50 % at 248 nm) and the trimer's temperature law (4.67 to 2.22 %). A band set in 1Dv0_2 (`Params1D.rates`, `--rate`) is a constant of the run in every cell, as in v0_2.

9 The system map

In the interior each cell is v0_2's 0-D run, so its energy flow is v0_2's map on the page after the gap figure. Figure 1 shows the gap itself at the reference point. The 248-nm share of each cell's own input is v0_2's value away from the wall on every mesh. It drops to nearly nothing in the wall cell, which narrows as the mesh refines, and on the finest mesh the cells beside it alternate about the interior value. At 500 ns, mid-pulse, the finest mesh shows the wall cell's electrons drained and held at the mean-energy ceiling, and n_e and T_e alternating beside it.

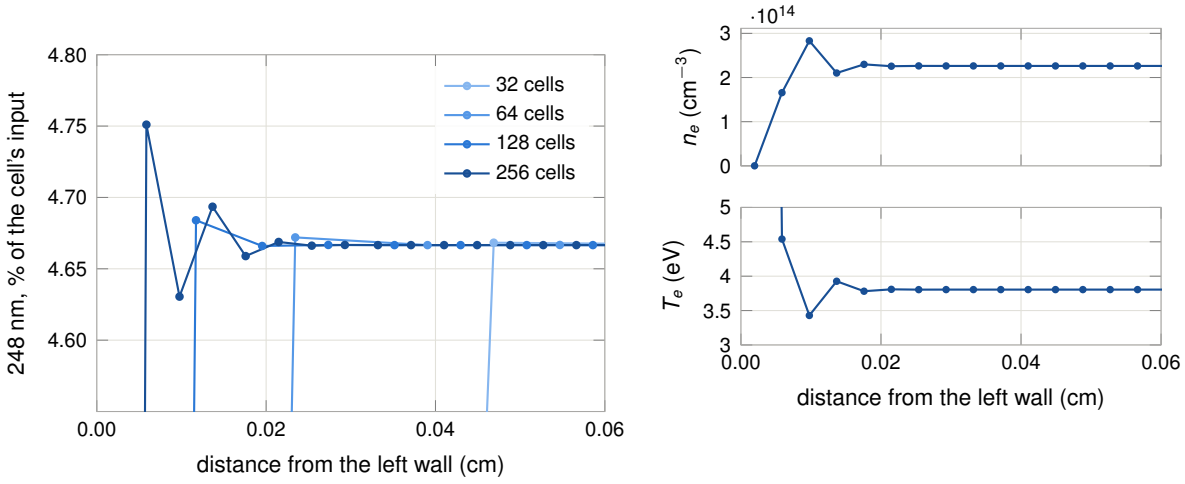
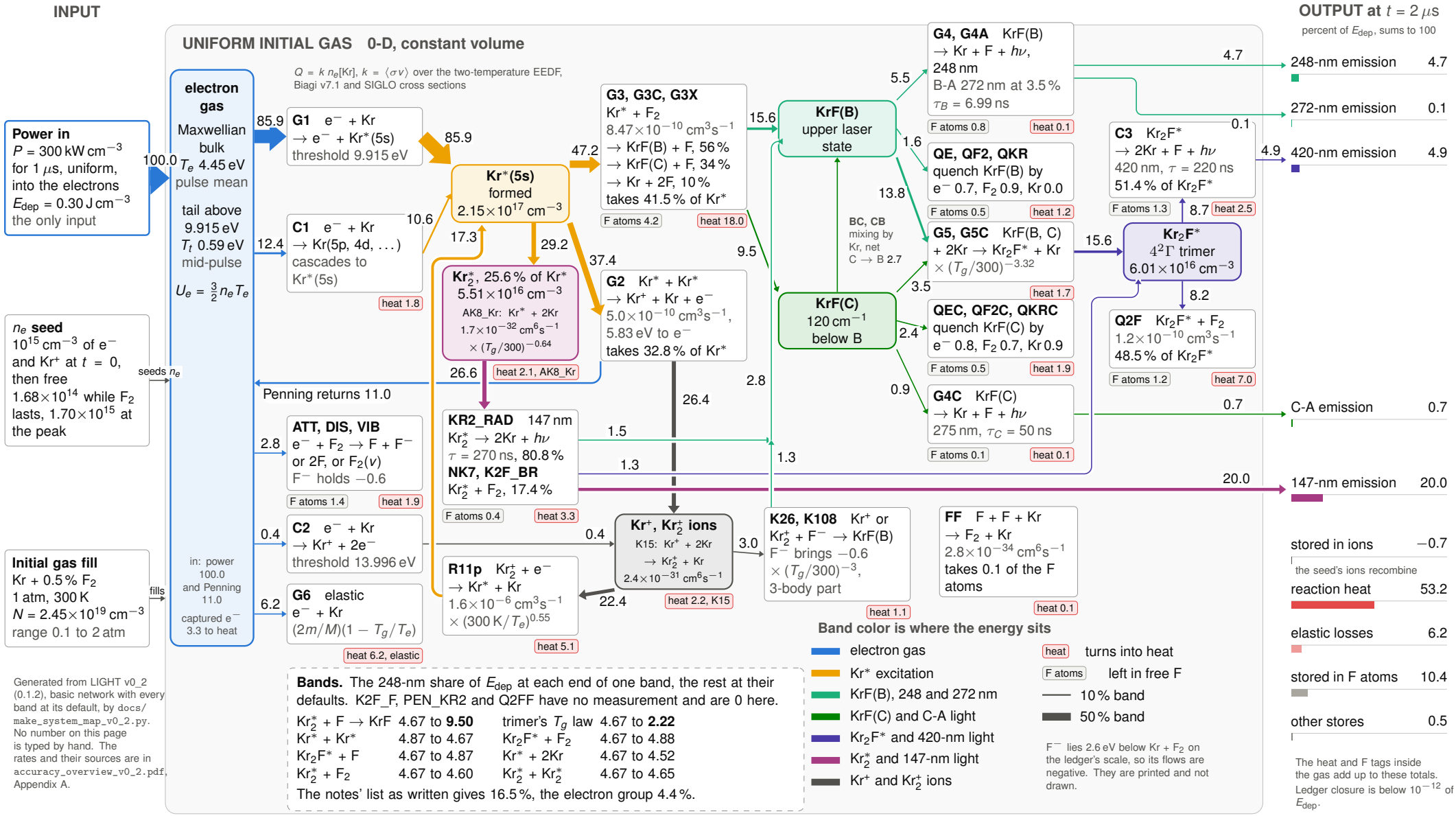


Figure 1: The end of the gap at v0_2's reference point, which is symmetric about its centre. Left, the 248-nm share of each cell's own input by the end of the run, on each mesh, about the interior's value. The wall cell's own share, near zero, lies below the axis. Right, n_e and T_e at 500 ns on 256 cells, where the wall cell's T_e at the ceiling lies above the axis. v0_2's 0-D share is 4.66672 %.

LIGHT system map, v0_2: energy flow through the Kr/F_2 kinetics

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then free, every band at its default, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Kr* 98.3 %, to Kr+ 26.8 %, to KrF(B) and KrF(C) 23.8 %, heat 53.2 %, elastic losses 6.2 %, emission 30.4 %. Of the Kr* formed, 25.6 % forms Kr_2^* (15.1 % of the Kr* used before the F_2 burns out), and 80.8 % of that Kr_2^* radiates at 147 nm. Of the ions that recombine, 88.7 % meet an electron and return to Kr*, and 11.3 % meet F^- and form KrF(B). Of the KrF formed, 8.93×10^{16} cm^{-3}, 4.7 % came from ion-ion recombination and 5.4 % from $Kr_2^* + F_2$. Of it 19.6 % radiates at 248 nm, 0.7 % at 272 nm and 3.3 % in the C-A band, 62.0 % forms the trimer, and electrons, F_2 and Kr quench 5.4 %, 5.7 % and 3.4 %. At the end of the run T_g = 649 K and p = 2.18 atm, and F_2 fell below 10 % of the fill at 640 ns.

10 The energy ladder

1Dv0_2 runs v0_2's network in every cell, so its energy ladder, Figure 2, is v0_2's. It places every species at its formation energy, the energy above ground-state Kr and F_2 at rest that Appendix A lists, and draws the rows that move energy between them, as the ladder view of the website's excimer simulation does. KrF(B) lies 5.54 eV below the ion pair $Kr^+ + F^-$, the measured $D_e(B)$, and 5.06 eV above Kr + F. The 248-nm photon carries 5.00 eV of that and leaves KrF(X) only 0.06 eV above Kr + F. The harpoon G3 heats the gas by 3.26 eV for each KrF(B) it makes, the ion-ion recombination K26 by 5.54 eV and the trimer's formation by 0.58 eV. Kr_2F^* emits 2.95 eV at 420 nm and leaves 1.53 eV of lower-state repulsion. On the Kr_2^* route, which is basic in v0_2, Kr_2^* emits 8.43 eV at 147 nm and leaves 0.77 eV, or meets F_2 and makes KrF(B) through NK7 or Kr_2F^* through K2F_BR. The 1-D layer adds no level and no row. It moves the charged species and the electrons' energy between the cells and into the walls.

11 Where 1Dv0_2 is not yet reliable

- Everything that limits v0_2 limits 1Dv0_2, above all Kr_2^* + F, which has no measurement, and the trimer's rate above 425 K (the v0_2 overview, Section 10).
- Nothing measured tests the 1-D layer. No spatially resolved Kr/F_2 measurement is among the sources held, and with uniform power the transport barely touches the interior.
- The wall cells are unresolved sheaths, and their share of any gap average is a property of the mesh. The cells beside them alternate about the interior and do not converge as the mesh refines. Quote the interior, or a gap average with its mesh.
- The transport's own inputs are not LIGHT's. The ion mobilities are Fluid1D's table, the electron transport a Maxwellian at the mean energy, and the gas temperature it reads is the fill's. With a nonuniform power profile (`--profile`) these would matter.
- $Kr_2^+ + e^-$ above 30 eV runs at its 30-eV value, where v0_2 follows the power law (Section 4).
- The seed. At hundreds of kW cm^{-3} a seed below 10^{13} cm^{-3} does not break down, and the summary warns.
- First order in Δt and in Δx, and the mean electron energy held to 0.03 to 60 eV.
- No discharge. Fluid1D's circuit would be a drive module, which LIGHT does not carry, and it would need a field-driven EEDF where v0_2's tail is set by the given power.

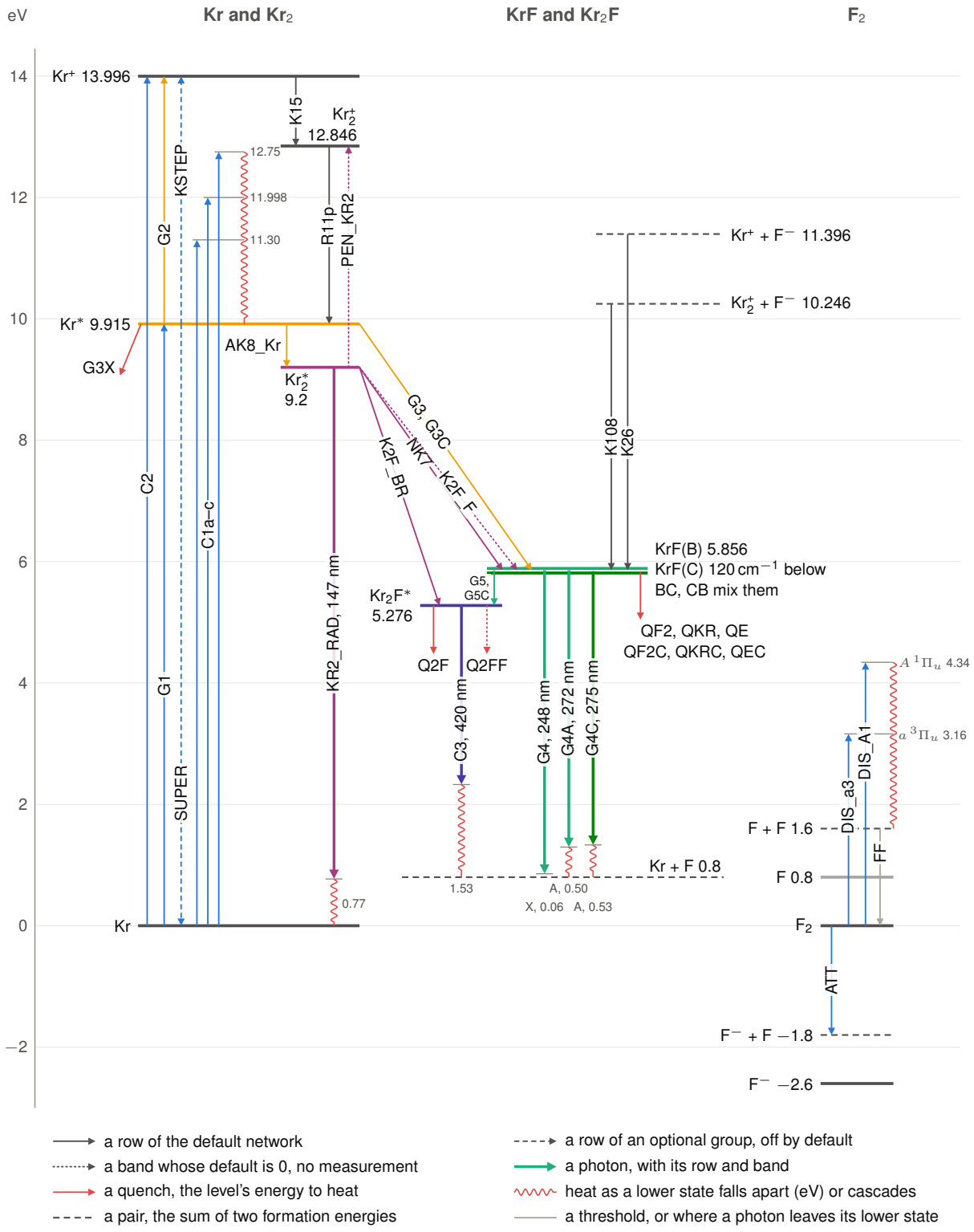


Figure 2: The energy ladder of LIGHT v0_2, version 0.1.2, in eV above ground-state Kr and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair, the sum of two. An arrow is a row, labelled with its id. It is solid in the default network and dotted for a band whose default is 0, Q2FF, K2F_F and PEN_KR2. The rows of the optional group `electron` are dashed. A photon's arrow drops by its quantum, and the red squiggle below it is the heat its lower state gives the gas as it falls apart. A red stub is a quench. The energy a row releases heats the gas, except a photon's, which leaves, and that of G2, PEN_KR2 and SUPER, which goes to the electrons. KrF(B) and KrF(C) lie 120 cm^{-1} apart and are drawn as a pair of bars that every B and C row meets. The rows G6, VIB1, VIB2, VIB3 and VIB4 move nothing between levels and are not drawn.

A Reaction rates

The rates are v0_2's, and 1Dv0_2 carries its files byte for byte, so these tables are the v0_2 overview's. Rates are the values at 300 K. The column T gives the law applied to them. Electron-impact rows use $\langle \sigma v \rangle$ over the EEDF, and Section A.3 lists their Maxwellian values. A row that reads a band prints the value it runs by default and the band's name, and Section A.4 lists the bands. Each row's full citation is in `1Dv0_2/network.py`.

A.1 The basic network, 44 rows

8 of them read a band.

row	reaction	rate	T	source
G1	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1, the whole 5s manifold at 9.915 eV
C1a	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1 5p group at 11.30 eV, prompt cascade to 5s
C1b	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1 d+p group at 11.998 eV, as C1a
C1c	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1 higher group at 12.75 eV, as C1a
C2	$e^- + Kr \rightarrow 2e^- + Kr^+$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1 ionization at 13.996 eV
G6	e^- + Kr, elastic	energy loss $(2m/M)\langle \sigma_m v \varepsilon \rangle$	T_e, T_g	Biagi v7.1 momentum transfer
ATT	$e^- + F_2 \rightarrow F^- + F$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO F_2 set
DIS_a3	$e^- + F_2 \rightarrow e^- + 2F$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO $a\,^3\Pi_u$, repulsive, so excitation dissociates
DIS_A1	$e^- + F_2 \rightarrow e^- + 2F$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO $A\,^1\Pi_u$, as DIS_a3
VIB1	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 1
VIB2	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 2
VIB3	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 3
VIB4	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 4
G2	$2Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10} cm^{3}s^{-1}, band kr_pen		Kannari 1985 Table I Penning block, from Leichner and Ericson 1974. Band down to Werner's 1.5×10^{-10}, via Jancaitis 1983
G3	$Kr^* + F_2 \rightarrow KrF(B) + F$	4.73×10^{-10} cm^{3}s^{-1}		rate from Velazco, Kolts and Setser 1976 and Ueda 1984, KrF^* yield 0.90 from Velazco, Kolts and Setser 1976, B share from Yu and Setser 1990
G3C	$Kr^* + F_2 \rightarrow KrF(C) + F$	2.9×10^{-10} cm^{3}s^{-1}		as G3, with Yu and Setser's k_B/k_C of 1.6
G3X	$Kr^* + F_2 \rightarrow Kr + 2F$	8.47×10^{-11} cm^{3}s^{-1}		as G3, the KrF^*-free rest, assigned to Kr + 2F
K15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.4×10^{-31} cm^{6}s^{-1}	$(T_g/300)^{-0.75}$	the mean of three measurements, 2.4×10^{-31}. T_g law from Perez-Rios and Greene 2018
R11p	$e^- + Kr_2^+ \rightarrow Kr^* + Kr$	1.6×10^{-6} $(T_e/300\,\mathrm{K})^{-0.55}$ cm^{3}s^{-1}	$T_e^{-0.55}$	Shiu and Biondi 1977, 1.6×10^{-6} $(300\,\mathrm{K}/T_e)^{0.55}$, $\pm$ 10 %. Mikus 1978 within 8 %
K26	$Kr^+ + F^- \rightarrow KrF(B)$	2.95×10^{-6} cm^{3}s^{-1} at 1 atm, 300 K, unscreened	N, screening, $(T_g/300)^{-3}$ on the 3-body part	Kannari 1985, Flannery turnover with Debye screening, and $(T_g/300)^{-3}$ on its three-body part from Thomson theory. Measured with F^- only below 300 Torr, Jacob et al. 1978. No measurement near 1 atm

row	reaction	rate	T	source
K108	$Kr_2^+ + F^- \rightarrow KrF(B) + Kr$	2.95×10^{-6} cm^3s^{-1} at 1 atm, 300 K, unscreened	N, screening, $(T_g/300)^{-3}$ on the 3-body part	as K26
G4	$KrF(B) \rightarrow Kr + F + h\nu$	1.38×10^{8} s^{-1}		Kannari 1985 Table I, the 96.6 % B-X share of τ_B = 6.99 ns. Eden 1978 measured 6.8 $\pm$ 0.2 ns
G4A	$KrF(B) \rightarrow Kr + F + h\nu$	4.84×10^{6} s^{-1}		$A_{B\text{-}A}/A_{B\text{-}X}$ = 0.035, Dunning and Hay's calculation via Yu and Setser 1990. 272 nm is a theoretical position
G4C	$KrF(C) \rightarrow Kr + F + h\nu$	2.0×10^{7} s^{-1}		ML joint fit. Hakuta 1990 measured 60 $\pm$ 10 ns. Band at 275 nm, printed
BC	$KrF(B) + Kr \rightarrow KrF(C) + Kr$	1.05×10^{-11} cm^3s^{-1}		ML joint fit. Hakuta 1990's 1.2×10^{-10} is held out
CB	$KrF(C) + Kr \rightarrow KrF(B) + Kr$	9.55×10^{-12} cm^3s^{-1}	$1/K_{eq}(T_g)$	detailed balance of BC, with the joint fit's K_{eq} and Yu and Setser's measured 120 $\pm$ 50 cm^{-1} gap
G5	$KrF(B) + 2Kr \rightarrow Kr_2F^* + Kr$	1.9×10^{-30} cm^6s^{-1}	$(T_g/300)^{-3.32}$	ML joint fit. T_g law from Shui's calculation for 300-800 K, measured only to 425 K
G5C	$KrF(C) + 2Kr \rightarrow Kr_2F^* + Kr$	3.79×10^{-31} cm^6s^{-1}	$(T_g/300)^{-3.32}$	ML joint fit
QE	$KrF(B) + e^- \rightarrow Kr + F + e^-$	6.6×10^{-8} cm^3s^{-1}		Trainor and Jacob's measured product, re-read with this network's lifetime
QEC	$KrF(C) + e^- \rightarrow Kr + F + e^-$	6.6×10^{-8} cm^3s^{-1}		as QE
QF2	$KrF(B) + F_2 \rightarrow Kr + 3F$	4.75×10^{-10} cm^3s^{-1}		ML joint fit. Xu 1993 printed 3.0 $\pm$ 0.3×10^{-10} with fast mixing
QF2C	$KrF(C) + F_2 \rightarrow Kr + 3F$	3.04×10^{-10} cm^3s^{-1}		ML joint fit
QKR	$KrF(B) + Kr \rightarrow 2Kr + F$	1.0×10^{-14} cm^3s^{-1}		ML joint fit, at its lower bound
QKRC	$KrF(C) + Kr \rightarrow 2Kr + F$	8.37×10^{-13} cm^3s^{-1}		ML joint fit
C3	$Kr_2F^* \rightarrow 2Kr + F + h\nu$	4.5×10^{6} s^{-1}		Xu, Gadomski and Setser 1993, 220 $\pm$ 20 ns. Band at 420 nm, printed
Q2F	$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	1.2×10^{-10} cm^3s^{-1}, band kr2f_f2		Xu 1993 Table I, 1.2 $\pm$ 0.2×10^{-10}. Band up to Quigley and Hughes' 4.3×10^{-10}
Q2FF	$Kr_2F^* + F \rightarrow 2Kr + 2F$	0, band kr2f_f		no Kr measurement, so 0 unless a band run sets it, 0 to 1.8×10^{-10}. The top is Jancaitis 1983's Ar_2F^* analog
FF	$2F + Kr \rightarrow F_2 + Kr$	2.8×10^{-34} cm^6s^{-1}		Cohen and Bott 1976 evaluation, 2.8×10^{-34}, every rare gas given the Ar value. Held at its 300 K value
AK8_Kr	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	1.7×10^{-32} cm^6s^{-1}, band kr2_form	$(T_g/300)^{-0.64}$	Ueda 1984 Table V, 1.7 $\pm$ 0.2×10^{-32} at 0.5-1 atm. T_g law from Wieme and Lenaerts 1981, 300-451 K. Band up to Hughes 1977, 5.5×10^{-32}
KR2_RAD	$Kr_2^* \rightarrow 2Kr + h\nu$	3.7×10^{6} s^{-1}		Morikawa 1989, 270 $\pm$ 5 ns, and five other measurements
NK7	$Kr_2^* + F_2 \rightarrow KrF(B) + Kr + F$	1.05×10^{-10} cm^3s^{-1}, band kr2_f2, half		half of Quigley and Hughes 1978, 2.1 $\pm$ 0.2×10^{-10}. The even split is Kannari's. Band up to Chen and Payne's 5.0×10^{-10}
K2F_BR	$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	1.05×10^{-10} cm^3s^{-1}, band kr2_f2, half		the other half of Quigley and Hughes 1978

row	reaction	rate	T	source
K2F_F	Kr_2^* + F → KrF(B) + Kr	0, band kr2_f		no measurement, so 0 unless a band run sets it, 0 to 1×10^{-9}. Kannari 1985's 3.0×10^{-10} is an estimate
PEN_KR2	$2Kr_2^*$ → Kr_2^+ + 2Kr + e^-	0, band kr2_pen		no measurement, so 0 unless a band run sets it, 0 to 5.0×10^{-10}. Kannari 1985 took 5.0×10^{-10} from an unpublished 1973 SRI report

A.2 Optional group on the basic network, 2 rows

The group electron is off at the reference point.

row	reaction	rate	T	source
KSTEP	e^- + Kr^* → Kr^+ + $2e^-$	$1.56\times10^{-7}\ T_e^{0.71}\times e^{-4.09/T_e}$ cm^3s^{-1}	T_e	Chen R9 template, an estimate
SUPER	e^- + Kr^* → e^- + Kr	from G1 by detailed balance	T_e	detailed balance of G1, statistical weight ratio 1/12

A.3 Electron-impact rate coefficients

Maxwellian $\langle\sigma v\rangle$ in cm^3s^{-1} at the T_e of each column. The last column is the factor the two-temperature tail applies at the reference point's mid-pulse state in 0-D, T_e = 3.81 eV with the tail at 0.16 T_e. The rows below the rule give the energy an attached electron takes from the electron gas. The stage reads these rows on the same T_e nodes.

row	process	T_e (eV)					tail
		0.5	1	1.5	2	3	factor
G1	Kr, 5s manifold, 9.915 eV	1.24×10^{-17}	3.44×10^{-13}	1.11×10^{-11}	6.41×10^{-11}	3.78×10^{-10}	0.036
C1a	Kr, 5p group, 11.3 eV	5.1×10^{-19}	5.93×10^{-14}	2.97×10^{-12}	2.11×10^{-11}	1.48×10^{-10}	5.53×10^{-3}
C1b	Kr, d+p group, 11.998 eV	2.5×10^{-19}	5.84×10^{-14}	3.77×10^{-12}	3.08×10^{-11}	2.53×10^{-10}	1.81×10^{-3}
C1c	Kr, higher group, 12.75 eV	9.15×10^{-21}	7.73×10^{-15}	8.8×10^{-13}	1.02×10^{-11}	1.29×10^{-10}	1.48×10^{-4}
C2	Kr, ionization, 13.996 eV	8.06×10^{-21}	1.57×10^{-14}	2.28×10^{-12}	2.94×10^{-11}	4.14×10^{-10}	5.09×10^{-5}
ATT	F_2, attachment	6.58×10^{-9}	3.51×10^{-9}	2.25×10^{-9}	1.61×10^{-9}	9.75×10^{-10}	1
DIS_a3	F_2, $a\,^3\Pi_u$, 3.16 eV	3.01×10^{-12}	1.13×10^{-10}	3.95×10^{-10}	7.37×10^{-10}	1.35×10^{-9}	1
DIS_A1	F_2, $A\,^1\Pi_u$, 4.34 eV	2.06×10^{-13}	1.93×10^{-11}	8.88×10^{-11}	1.9×10^{-10}	4.02×10^{-10}	1
VIB1	F_2, v = 1, 0.111 eV	1.77×10^{-9}	1.22×10^{-9}	8.72×10^{-10}	6.61×10^{-10}	4.27×10^{-10}	1
VIB2	F_2, v = 2, 0.219 eV	2.01×10^{-9}	1.49×10^{-9}	1.09×10^{-9}	8.37×10^{-10}	5.47×10^{-10}	1
VIB3	F_2, v = 3, 0.324 eV	1.21×10^{-9}	9.75×10^{-10}	7.35×10^{-10}	5.71×10^{-10}	3.78×10^{-10}	1
VIB4	F_2, v = 4, 0.421 eV	3.83×10^{-10}	3.27×10^{-10}	2.51×10^{-10}	1.96×10^{-10}	1.31×10^{-10}	1
ATT	energy an attached e^- takes, eV	0.37	0.46	0.53	0.59	0.72	
	$\frac{3}{2}T_e$, for comparison	0.75	1.50	2.25	3.00	4.50	

A.4 The bands

`network.BANDS`. `Params1D.rates` sets a band, and `Params1D.trimer_tg_pow` the trimer law.

band	process	default	range	source
kr2_f	Kr_2^* + F → KrF(B) + Kr	0	0 to 1.0×10^{-9} cm^3s^{-1}	No measurement. Kannari's 3.0×10^{-10} traces to SRI MP-76-99, whose model leaves the reaction out (p.45) and whose one number is a conditional 3-10×10^{-10} for Ar_2^* + F (p.65), the band's top. Jancaitis 1983 has no such row

band	process	default	range	source
trimer_tg_pow	the trimer's $(T_g/300)^n$	-3.32	-3.32 to -0.50 exponent	Shui's calculation over 300-800 K, the default, measured to 425 K. Salesky and Kimura inferred -0.5 from pure-Kr efficiencies, a bound that never sets it. Params.trimer_tg_pow sets it
kr_pen	$Kr^* + Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10}	1.5×10^{-10} to 5.0×10^{-10} cm^3s^{-1}	Kannari 1985 gives 5.0×10^{-10} from Leichner and Ericson 1974, v0's and the default. Jancaitis's model carries 1.5×10^{-10} from Werner's thesis (1975)
kr2f_f2	$Kr_2F^* + F_2$	1.2×10^{-10}	1.2×10^{-10} to 4.3×10^{-10} cm^3s^{-1}	Xu 1993 measured $1.2 \pm 0.2\times10^{-10}$, the default, and lists 1.5, 1.8 and 4.3×10^{-10} before it. Jancaitis 1983 measured $1.8 \pm 0.2\times10^{-10}$
kr2f_f	$Kr_2F^* + F \rightarrow 2Kr + 2F$	0	0 to 1.8×10^{-10} cm^3s^{-1}	No Kr measurement. Jancaitis 1983 found F atoms quench Ar_2F^* at 0.5-1 times F_2 (p.153). With his $Kr_2F^* + F_2$, 1.8×10^{-10}, that analog gives 0.9-1.8×10^{-10} and sets the top
kr2_form	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$, 300 K	1.7×10^{-32}	1.7×10^{-32} to 5.5×10^{-32} cm^6s^{-1}	Ueda 1984 measured $1.7 \pm 0.2\times10^{-32}$ at 0.5-1 atm, the default. Low-pressure afterglows give 1.79 to 4.4×10^{-32} and Hughes 1977 5.5×10^{-32}
kr2_f2	$Kr_2^* + F_2$, both channels	2.1×10^{-10}	2.1×10^{-10} to 5.0×10^{-10} cm^3s^{-1}	Quigley and Hughes 1978 measured $2.1 \pm 0.2\times10^{-10}$, the default. SRI MP-76-99 gives 3.0×10^{-10} and Chen and Payne 1979 5.0×10^{-10}, beyond each other's errors
kr2_pen	$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	0	0 to 5.0×10^{-10} cm^3s^{-1}	No measurement. Kannari's 5.0×10^{-10} is Lorents and Olson's unpublished 1973 SRI report, which gives Ar_2^* the same value, and Jancaitis's model carries 1.0×10^{-10} from Werner et al. 1977

A.5 Formation energies

Energies above ground-state Kr and F_2 at rest. Every reaction's energy release is the difference of these numbers, so no row can make or lose energy. c_v is the heat capacity per particle in units of k_B.

species	E_{form} (eV)	c_v	what sets it
Kr	0	1.5	ground state
F_2	0	2.5	ground state
F	0.8	1.5	half the F_2 bond, D_0 = 1.60 eV
Kr^*	9.915	1.5	Biagi's 5s threshold, the 5s[3/2]2 level
Kr^+	13.996	1.5	Biagi's ionization threshold
KrF(B)	5.856	2.5	$D_e(B)$ = 5.54 eV below the ion pair, Tellinghuisen 1976
KrF(C)	5.84112	2.5	120 cm^{-1} below B, Yu and Setser 1990
Kr_2F^*	5.276	3	0.58 eV below KrF^* + Kr, calculated by Huestis et al. 1984
Kr_2^*	9.2	2.5	the 147-nm excimer, an estimated well
Kr_2^+	12.846	2.5	IP(Kr) less $D_0(Kr_2^+)$ = 1.15 eV, measured by photoionization, Ng et al. 1977
F^-	-2.6	1.5	E_{form}(F) less the electron affinity, 3.40 eV